

## **Key Findings and Recommendations**

The objective of this project was to identify customers at risk of churning and determine the factors contributing to customer attrition. Through exploratory data analysis and machine learning modeling, several important patterns were identified.

### **Key Findings**

- Customer churn is influenced by a combination of demographic, behavioral, and financial factors.
- Older customers showed a higher likelihood of leaving the bank compared to younger customers.
- Inactive customers were significantly more likely to churn than active customers.
- Geographic location appeared to impact churn behavior, indicating regional differences in customer retention.
- Customers with fewer banking products tended to have a higher probability of leaving.
- Machine learning models successfully identified patterns associated with churn, allowing high-risk customers to be detected before they leave.

### **Business Problems Identified**

1. **Customer Inactivity**
  - A large proportion of churned customers were inactive members.
  - Low engagement may lead customers to seek alternative banking services.
2. **High-Risk Customer Segments**
  - Certain age groups and customer profiles exhibited higher churn rates.
  - Without targeted interventions, these groups are more likely to leave.
3. **Regional Churn Variations**
  - Churn rates differed across geographic regions.
  - This suggests that customer needs and satisfaction levels vary by location.
4. **Limited Product Adoption**
  - Customers with fewer products demonstrated lower loyalty and engagement.
  - Limited interaction with the bank increases the likelihood of churn.

## **Recommendations**

- Develop targeted retention campaigns for customers identified as high-risk.
- Increase engagement initiatives for inactive customers through personalized communication and loyalty programs.
- Encourage customers to adopt additional banking products and services to strengthen customer relationships.
- Conduct region-specific analyses to better understand local customer needs and improve service delivery.
- Integrate the predictive model into business operations to proactively identify customers likely to churn and intervene before customer loss occurs.

## **Conclusion**

This project demonstrates how machine learning can be used to support customer retention strategies. By identifying customers at risk of churning and understanding the factors driving their decisions, organizations can take proactive measures to improve customer satisfaction, reduce attrition, and increase long-term profitability.